Core Concepts of Large Language Models (LLMs)

1. Tokenization

- LLMs process text by breaking it into smaller units called tokens (words, subwords, or characters).
- Example: "ChatGPT is powerful" -> ['Chat', 'G', 'PT', 'is', 'powerful'].

2. Embeddings

- Each token is converted into a high-dimensional vector that captures semantic meaning.
- Similar words have embeddings that are close together in vector space.

3. Neural Architecture (Transformer)

- LLMs use a Transformer architecture based on self-attention mechanisms.
- Self-attention lets the model weigh the importance of each word relative to others in a sentence.

4. Attention Mechanism

- Attention helps the model focus on relevant parts of the input when generating outputs.
- It computes weighted averages of embeddings based on contextual relationships.

5. Training Process

- LLMs are trained on massive text datasets to predict the next token in a sequence.
- Loss functions (like cross-entropy) measure how well predictions match actual tokens.

6. Fine-tuning & Instruction Tuning

- Fine-tuning adapts a pretrained model to specific tasks or domains.
- Instruction tuning helps models better follow human instructions.

7. Context Window

- Refers to how much text the model can "see" at once.
- A larger context window enables understanding of longer conversations or documents.

8. Prompt Engineering

- Crafting input text (prompts) to guide the model's responses effectively.
- Example: "Summarize this paragraph in two sentences."

9. Inference & Sampling

- During generation, models use sampling techniques (e.g., greedy, top-k, nucleus) to produce outputs.
- Temperature controls randomness - higher values make responses more creative.

10. Limitations & Bias

- LLMs can reflect biases present in training data.
- They may also generate incorrect or misleading information (hallucination).